Note to Co-authors: Cost-Side Robustness Checks

This note summarizes the first round of exchange-rate identification robustness checks for the cost-side regression. The main point is that the imported-parts cost channel survives the stricter-sample and country-omission checks, and the foreign-assembled placebo looks reassuring, but the future-exchange-rate placebo remains active even after the sample-construction fix, the 2025 exchange-rate backfill, and the addition of the contemporaneous exposure-weighted exchange-rate term in the same regression. The country parser has also been repaired and shared with post-estimation code: raw `UK`/`GB`/`Great Britain` labels canonicalize to `United Kingdom`, and `Korea`/`South Korea` canonicalize to `Korea`. Direct product-level source-country exchange-rate joins now use the full normalized exchange-rate table; the top parts-import-country list is used only for the aggregate trade-weighted RER control. So the strongest version of the identification claim should not treat either the levels or first-difference future-shock placebo as a clean pass.

Panel and samples

The scripted panel builder now produces:

- a domestic non-switcher panel with 478 rows and 102 make-models,
- a domestic strict sample with 444 rows and 99 make-models, and
- a domestic single-row sample with 393 rows and 92 make-models.

The baseline domestic levels regression uses 376 observations. The baseline domestic first-difference regression uses 256 observations.

For the future-RER placebo specifically, the IMF-based 2025 exchange-rate backfill makes direct RER_{t+1} available for all 376 domestic current-sample rows. The future-RER first-difference sample is now the full 256-row current first-difference sample because $\Delta \log(RER_{t+1})$ is constructed directly as $\log(RER_{t+1}) - \log(RER_t)$ from the current row's source country rather than from the next observed make-model row.

Placebo regressions

Table 1 reports the levels placebo results. Table 2 reports the first-difference placebo results. In both tables, column (3) is the conditional current-plus-future exchange-rate specification: it tests whether $\rho_{j,t-1} \log(RER_{j,t+1})$ predicts current recovered costs after controlling for $\rho_{j,t-1} \log(RER_{jt})$. Column (4) is the foreign-assembled placebo.

Table 1: Exchange-rate placebo regressions (levels)

Dependent Variable:	ln_costs			
Model:	(1)	(2)	(3)	(4)
Variables				
ln(size)	0.5452***	0.5044***	0.5046***	-0.1112
	(0.2005)	(0.1904)	(0.1908)	(0.1976)
ln(weight)	0.3020*	0.3832**	0.3823**	0.3867**
	(0.1746)	(0.1593)	(0.1618)	(0.1549)
ln(hp)	0.1884**	0.1948**	0.1946**	0.2554***
	(0.0825)	(0.0781)	(0.0789)	(0.0532)
ln(mpg)	-0.0960	-0.0685	-0.0688	-0.0930
	(0.0759)	(0.0782)	(0.0780)	(0.0651)
$\rho_{j,t-1} \times \log(RER_{jt})$	0.6009***		0.0177	-0.0495
	(0.2286)		(0.2119)	(0.0772)
$\rho_{j,t-1} \times \log(RER_{j,t+1})$		0.7349***	0.7221***	
		(0.2223)	(0.2171)	
Fixed-effects				
make_model	Yes	Yes	Yes	Yes
year	Yes	Yes	Yes	Yes
Fit statistics				
Observations	376	376	376	882
R^2	0.97894	0.97968	0.97968	0.99152
Within R^2	0.18104	0.20986	0.20988	0.18728

Clustered (make_model) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 2: Exchange-rate placebo regressions (first differences)

Dependent Variable:	ln_costs			
Model:	(1)	(2)	(3)	(4)
Variables				
Δ ln(size)	0.5071*	0.5359**	0.5218**	-0.0119
	(0.2564)	(0.2563)	(0.2546)	(0.1926)
Δ ln(weight)	0.3996*	0.4186*	0.4042*	0.2900**
	(0.2220)	(0.2223)	(0.2227)	(0.1303)
Δ ln(hp)	0.2127**	0.2057**	0.2053**	0.2855***
	(0.0999)	(0.0970)	(0.0981)	(0.0569)
Δ ln(mpg)	-0.0476	-0.0597	-0.0587	-0.1342*
	(0.0695)	(0.0765)	(0.0745)	(0.0787)
$\rho_{j,t-1} \times \Delta \log(RER_{jt})$	0.4605**		0.2470	-0.0457
	(0.2122)		(0.2298)	(0.0805)
$\rho_{j,t-1} \times \Delta \log(RER_{j,t+1})$		0.5558**	0.4652**	
		(0.2111)	(0.2258)	
Fixed-effects				
year	Yes	Yes	Yes	Yes
Fit statistics				
Observations	256	256	256	590
R^2	0.14424	0.15158	0.15371	0.32981
Within R^2	0.11399	0.12159	0.12380	0.23399

Clustered (make_model) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Stricter exposure-sample checks

Table 3 re-estimates the baseline pass-through specification on the baseline non-switcher, strict, and single-row samples. Table 4 does the same for the elasticity-interaction specification used by the counterfactual pipeline.

Table 3: Baseline pass-through robustness across stricter exposure samples

Dependent Variable:	ln_costs					
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
ln(size)	0.5452***	0.5570***	0.4424**	0.5071*	0.4019*	0.1416
	(0.2005)	(0.1887)	(0.2023)	(0.2564)	(0.2205)	(0.1244)
ln(weight)	0.3020*	0.3264	0.3615	0.3996*	0.5236	0.4557
	(0.1746)	(0.2018)	(0.2220)	(0.2220)	(0.4033)	(0.5165)
ln(hp)	0.1884**	0.1715**	0.1439*	0.2127**	0.1659**	0.1685**
	(0.0825)	(0.0791)	(0.0765)	(0.0999)	(0.0756)	(0.0842)
ln(mpg)	-0.0960	-0.0802	-0.0546	-0.0476	-0.0331	-0.0126
	(0.0759)	(0.0774)	(0.0787)	(0.0695)	(0.0598)	(0.0553)
$\rho_{j,t-1} \times \log(RER_{jt})$	0.6009***	0.6841***	0.7553***	0.4605**	0.7754***	0.8217**
	(0.2286)	(0.2315)	(0.2627)	(0.2122)	(0.2871)	(0.3148)
Fixed-effects						
make_model	Yes	Yes	Yes			
year	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	376	345	301	256	225	192
R^2	0.97894	0.97784	0.97964	0.14424	0.15596	0.13792
Within R^2	0.18104	0.18411	0.17531	0.11399	0.12169	0.11589

Clustered (make_model) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 4: Elasticity-interaction robustness across stricter exposure samples

Dependent Variable:	ln_costs					
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
ln(size)	0.4892**	0.4995***	0.3811*	0.4969**	0.3995*	0.1496
	(0.1940)	(0.1820)	(0.1965)	(0.2471)	(0.2123)	(0.1299)
ln(weight)	0.2775	0.2983	0.3285	0.4000*	0.5393	0.5115
	(0.1708)	(0.2004)	(0.2217)	(0.2209)	(0.4011)	(0.5104)
ln(hp)	0.2025**	0.1904**	0.1724**	0.2024**	0.1512**	0.1469*
	(0.0786)	(0.0751)	(0.0680)	(0.0964)	(0.0755)	(0.0851)
ln(mpg)	-0.0729	-0.0503	-0.0089	-0.0289	-0.0113	0.0109
	(0.0727)	(0.0725)	(0.0716)	(0.0700)	(0.0586)	(0.0530)
$\rho_{j,t-1} \times \log(RER_{jt})$	4.532***	4.904***	6.202***	4.047*	4.769**	5.721***
	(1.332)	(1.420)	(1.432)	(2.049)	(2.136)	(2.149)
$\rho_{j,t-1} \times \log\lvert\varepsilon_{j,t-1}\rvert \times \log(RER_{jt})$	-2.041***	-2.191***	-2.806***	-1.835*	-2.031*	-2.475**
	(0.6783)	(0.7304)	(0.7518)	(1.052)	(1.080)	(1.094)
Fixed-effects						
make_model	Yes	Yes	Yes			
year	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	376	345	301	256	225	192
R^2	0.98004	0.97915	0.98175	0.15913	0.17416	0.16764
Within R^2	0.22364	0.23246	0.26089	0.12941	0.14062	0.14637

Clustered (make_model) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Adding a plain exchange-rate main effect

Table 5 adds an unrestricted exchange-rate main effect alongside the imported-parts exposure interaction on the baseline non-switcher sample.

Leave-one-country-out

Table 6 summarizes the leave-one-country-out baseline regressions. Omitted countries are now generated dynamically from every primary source country observed in the baseline regression frame. China parses correctly and appears in the broader domestic panel, but it is not present in the current domestic non-switcher regression frame.

Takeaways

- The foreign-assembled placebo is close to zero in both levels and first differences.

Table 5: Baseline pass-through robustness with an exchange-rate main effect

Dependent Variable:	ln_costs	
Model:	(1)	(2)
Variables		
$\log(RER_{jt})$	-0.2227*	-0.2302
	(0.1198)	(0.2053)
ln(size)	0.5406**	0.4952*
	(0.2058)	(0.2701)
ln(weight)	0.3356*	0.3877*
	(0.1693)	(0.2315)
ln(hp)	0.1954**	0.2198**
	(0.0849)	(0.1009)
ln(mpg)	-0.0821	-0.0429
	(0.0768)	(0.0684)
$\rho_{j,t-1} \times \log(RER_{jt})$	1.237***	1.190*
	(0.4165)	(0.6523)
Fixed-effects		
make_model	Yes	
year	Yes	Yes
Fit statistics		
Observations	376	256
R^2	0.97926	0.14829
Within R^2	0.19338	0.11818

Clustered (make_model) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

Table 6: Leave-one-country-out baseline pass-through regressions

Omitted country	Levels coef.	Levels N	FD coef.	FD N
Full sample	0.601 (0.229)	376	0.461 (0.212)	256
Mexico	1.263 (0.494)	201	1.188 (0.489)	144
Japan	0.617 (0.506)	229	0.024 (0.435)	143
Korea	0.631 (0.228)	343	0.506 (0.221)	239
Germany	0.608 (0.224)	357	0.459 (0.218)	243
Spain	0.600 (0.229)	374	0.450 (0.214)	255

- The conditional current-plus-future placebo is the main timing diagnostic because exchange rates are persistent. It asks whether the future shock predicts current recovered costs after conditioning on the contemporaneous exposure-weighted exchange-rate shock.
- The future-exchange-rate placebo remains active even after fixing the sample-construction bug and recovering the 2024 rows through the 2025 exchange-rate backfill. In the joint levels regression, the current term is 0.018 (SE 0.212) and the future term is 0.722 (SE 0.217).
- In the joint first-difference regression, the current term is 0.247 (SE 0.230) and the future term is 0.465 (SE 0.226). The future term remains positive, though less precisely estimated than in levels. So the future-shock placebo is not a clean pass in first differences either.
- The baseline pass-through coefficient remains positive and somewhat larger on the stricter samples.
- When we add a plain exchange-rate main effect, the exposure interaction remains positive and becomes larger: 1.237 (SE 0.416) in levels and 1.190 (SE 0.652) in first differences. The plain exchange-rate term itself is negative and only marginally precise in levels.
- The elasticity-interaction specification is stable in sign across the stricter samples: the main `rho x log(RER)` term stays positive and the `rho x log|elasticity| x log(RER)` term stays negative.
- Omitting Mexico, Japan, Korea, Germany, Spain, or Sweden does not flip the sign of the baseline coefficient, though Mexico matters most for magnitudes.

My read is that these checks do not kill the main imported-parts channel, but they do narrow the cleanest identification claim. The paper should keep the foreign placebo, treat the future-shock placebo as a real concern rather than a resolved one, and avoid overselling either the levels or the first-difference placebo as fully dispositive.